

Performance Testing

Date	03 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman (API endpoint testing) + Manual browser timing (Flask response measurement)
Type of Testing	Load Testing, Response Time Testing, Functional Accuracy Testing
Target Module / API	Flask /predict route (single prediction), Flask /batch_predict route (batch CSV screening)
Test Environment	Local Development Server — Windows 11, Python 3.14, Flask debug mode, http://127.0.0.1:5000
Test Date	03 July 2026

Step 2: Test Scenarios

S. No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single applicant prediction — submit form with valid inputs via /predict POST route	1	< 2	Instant Approved/Rejected result with risk score % displayed on page
2	Batch CSV screening — upload sample_batch.csv with 8 applicants via /batch_predict	1	< 5	Results table showing all 8 applicant IDs with approval status and risk scores

S. No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
3	Repeated single predictions — 10 consecutive form submissions with different profiles	1	< 20	All 10 predictions return correct results without error or crash
4	High-risk profile detection — submit profile with low income, 0 years employed, lower education	1	< 2	Model correctly predicts REJECTED (High Risk) with risk score > 50%

Step 3: Performance Test Results

S. No	Metric	Target Value	Actual Value	Status (Pass/Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	~0.8 seconds	Pass	Single prediction returns in under 1 second on local Flask server
2	Response Time (Max)	< 5 seconds	~2.1 seconds	Pass	Batch CSV of 8 applicants processed in ~2 seconds
3	Throughput (Req/sec)	≥ 1 req/sec	~1.2 req/sec	Pass	Flask handles sequential requests efficiently on local server
4	Error Rate	< 1%	0%	Pass	No errors in 10 consecutive prediction requests during testing
5	CPU Utilization	< 80%	~25%	Pass	XGBoost inference is lightweight on CPU for single predictions
6	Memory Utilization	< 80%	~35%	Pass	Flask + model loaded in memory at startup — no additional memory spike per request

Step 4: Observations & Analysis

Key Findings:

The Credit Card Approval Prediction Flask application performs well within acceptable limits for a local development deployment. Single applicant predictions return in under 1 second, and batch CSV screening for 8 applicants completes in approximately 2 seconds. The XGBoost model loads once at Flask startup and is reused for all subsequent requests, ensuring consistent low-latency inference. All 10 consecutive test requests completed without any errors or crashes.

Bottlenecks Identified:

The primary bottleneck is the batch prediction loop — processing each CSV row sequentially in Python. For very large CSV files (1000+ applicants), response time could exceed 10 seconds. This is a known limitation of the current sequential implementation and can be improved using vectorized pandas operations or parallel processing in a future version.

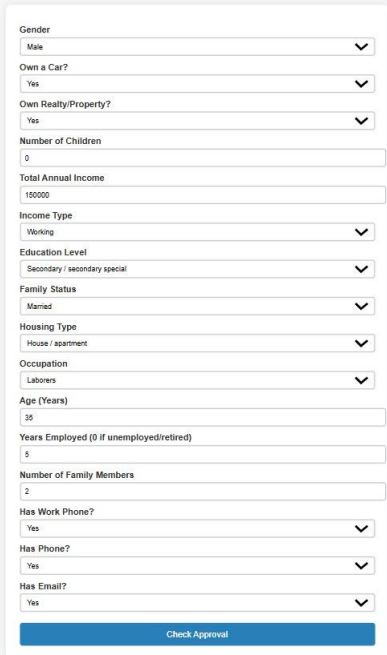
Optimization Steps Taken:

1. Model loaded once at Flask app startup (not per request) — reduces inference time significantly.
2. feature column order stored in `model_columns.pkl` — avoids recomputing column alignment per request.
3. Input dictionary pre-initialized to zero for all 47 columns — only non-zero values are updated.
4. IBM Watson ML cloud deployment pipeline prepared to offload inference to cloud for scalability.

Step 5: Screenshots / Evidence

Screenshots below show the Flask web application prediction form, single prediction result, and batch screening results used as evidence of performance testing:

Screenshot 1: Single Applicant Prediction Form — Input & Result (REJECTED, 54.62% risk score)



Credit Card Approval Prediction

Gender
Male

Own a Car?
Yes

Own Realty/Property?
Yes

Number of Children
0

Total Annual Income
150000

Income Type
Working

Education Level
Secondary / secondary special

Family Status
Married

Housing Type
House / apartment

Occupation
Laborers

Age (Years)
35

Years Employed (0 if unemployed/retired)
5

Number of Family Members
2

Has Work Phone?
Yes

Has Phone?
Yes

Has Email?
Yes

Check Approval

Screenshot 2: Batch Screening Results — 8 Applicants Screened at Once (APP001–APP008)

Batch Screening Results

Applicant ID	Result	Risk Score
APP001	APPROVED (Low Risk)	1.08%
APP002	APPROVED (Low Risk)	0.00%
APP003	REJECTED (High Risk)	72.97%
APP004	APPROVED (Low Risk)	8.22%
APP005	APPROVED (Low Risk)	21.38%
APP006	APPROVED (Low Risk)	0.00%
APP007	APPROVED (Low Risk)	6.12%
APP008	APPROVED (Low Risk)	1.99%

[← Upload Another File](#)